∞-brush: Controllable Large Image Synthesis with Diffusion Models in Infinite Dimensions

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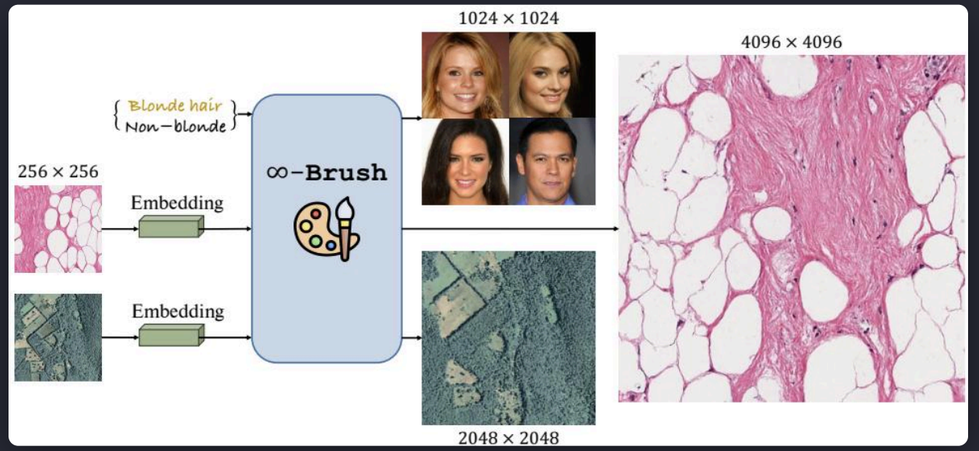
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Problem Statement

- Current diffusion models limited to fixed training resolutions
- Cannot exceed 1024×1024 without fidelity loss (e.g., SDXL)
- Patch-based methods fail to capture long-range spatial relationships
- Heavy reliance on local information during generation
- No existing model enables controllable generation at 4096×4096
- Computational demands scale quadratically with resolution

Our Solution

- Introduce ∞-Brush: First conditional diffusion model in function space
- Generates images at arbitrary resolutions up to 4096×4096
- Key innovation: Cross-attention neural operator
- Enables conditioning in infinite-dimensional space
- Training efficiency: Only 0.4% pixel subsets per image
- $65,536$ pixels instead of 16 million for 4096×4096

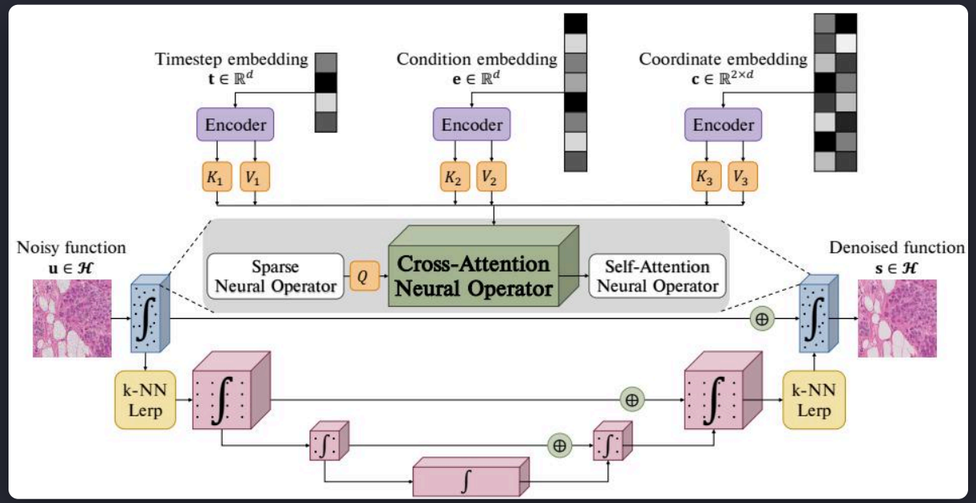


Related Work Limitations

- **Finite-dimensional models** (SDXL, Matryoshka)
- Fixed resolution constraint: Cannot exceed training resolution
- Quadratic computational cost with resolution
- **Patch-based methods** (MultiDiffusion, Graikos et al.)
- Captures local details but loses global structure
- No conditioning for controllable generation
- **Infinite-dimensional models** (∞-Diff)
- Enables arbitrary resolution but no conditional control

Core Innovation

- **Cross-attention neural operator**
- Replaces vanilla attention in function space
- Complexity: $O((N + \Sigma N_l)d^2)$ vs. $O(N^2 d)$
- **Conditional diffusion process**
- Forward: $Q(u_t|u_0) = N(u_t; \sqrt{\bar{\alpha}_t}Au_0, (1-\bar{\alpha}_t)ACA^T)$
- **Training efficiency**
- 0.4% pixel subsets per image ($65,536$ vs. 16M pixels)

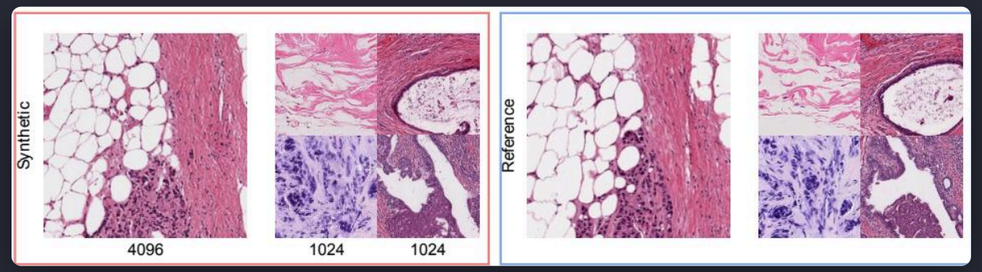


Experimental Setup

- **Datasets**
- CelebA-HQ (`1024 × 1024`): 30k images + hair color conditioning
- TCGA-BRCA (`4096 × 4096`): 57k histopathology patches
- NAIP satellite (`1024 × 1024`): 35k images
- **Evaluation metrics**
- CLIP FID: Global structure quality
- Crop FID: Local detail fidelity
- **Baselines**
- SDXL, Graikos et al. (patch-based), ∞-Diff

Results: Global Structure

- **TCGA-BRCA `4096 × 4096`**
- ∞-Brush: CLIP FID = 3.81 (vs. 7.43 for patch-based)
- Takeaway: Superior global structure preservation
- **NAIP `1024 × 1024`**
- ∞-Brush: CLIP FID = 6.32 (vs. 10.90 for SDXL)
- **SDXL infeasible at `4096 × 4096`**
- Requires full-resolution training (impossible on standard hardware)



Results: Efficiency & Trade-offs

- **Cross-attention operator critical**
- Ablation: removing it degrades both global structure and fine detail
- **vs. SDXL**
- Better global structure; modest trade-off in fine details
- Substantially lower memory and training time; SDXL infeasible at 4096×4096
- **vs. patch-based methods**
- Better global coherence at very large resolutions; modest trade-off in local detail
- Constant training cost across resolutions vs. growing patch-based overhead
- **vs. ∞-Diff**
- Adds conditional control with better global structure quality
- Trains on far smaller pixel subsets, cutting memory and compute
- Takeaway: Consistent scalability across resolutions

Key Contributions

- First conditional diffusion model in function space
- Enables controllable generation at arbitrary resolutions
- Cross-attention neural operator for infinite dimensions
- Linear complexity vs. quadratic for vanilla attention
- Training on `0.4%` pixel subsets
- Achieves `4096 × 4096` synthesis with 10x lower memory

Limitations & Future Work

- **Limitations**
- Fine details slightly worse than patch-based methods
- Smaller parameter count (450M vs. 860M for [10])
- **Future work**
- Design local neural operators for fine details
- Transfer knowledge from finite-dimensional models
- Scale to higher resolutions (8192×8192)

Conclusion

- ∞-Brush overcomes resolution limits of existing models
- First to enable controllable 4096×4096 synthesis
- Key innovation: Cross-attention neural operator
- Maintains global structure while preserving details
- Computational efficiency:
- 12 hr/epoch vs. 140 hr for SDXL at 1024×1024
- Constant training cost across resolutions